0 · COURSE & DESIGN PROCESS

Grade: labs 25, midterm 25, final 25, project 25, +5 participation; HW ungraded but exams cover HW, labs, lectures.
4 phases: Research → Design (schedule, spreadsheet, block diagram, sim) → Implementation (checklist, schematics, reviews before ordering) → Assembly & test. Software planned from start; stub functions.
Design review: 1-3 reviewers + moderator; docs 3-4 days ahead; find issues, don't solve; sign-off before layout; ROI >20×.
Test plan: raw board (Vcc-GND infinite Ω) → loaded board → power with current limit, check heat/odor.
PCB faults: 75-90% mechanical (holes, footprints, clearances).
Moore: 2× transistors ≈18 mo (reading says 2 yr). Leakage ≈ switching power by 2001 (Dennard scaling ended); fixes UTB-SOI, 3-D FinFET.
Analog computers: op-amps add/sub/mult/integrate, ~5%; revival as analog-approx + digital-refine hybrids.

1 · CLOCKS & PULSES

f=1/T (1 ns=1 GHz, 1 µs=1 MHz). Duty=$t_{hi}/(t_{hi}+t_{lo})$. Pulse width @50%; rise 10→90%, fall 90→10%. Overshoot above rail, undershoot below GND, ringing = damped osc. (mismatch, bad bypass, inductance). Prop. delay ~2 ns/ft on PCB; all logic must fit in one period. Clock skew = arrival diff. across parts, cuts margin.

2 · PCB DESIGN

Flow: schematic → label (R C Q U J) → place → route → review. R=ρL/A, A=w·t; drop=IR, P=I²R; 1 mil=.001 in=.00254 cm.
Planes: low-Z power/return; bypass caps ~1 per 1-2 ICs. Split analog/digital planes joined at **one point** (avoid ground loops). Sockets, test points, probe room. Terminate if len ≥ $t_R/(2\cdot t_{PR})$, t_{PR}≈150 ps/in (0.5 ns→1.67 in); parallel R=Z_T, Thévenin R1‖R2=Z_T. Fan-out TTL≈10; buffer: 74x245 data (bidir), 74x244 addr. CMOS $P \propto V^2 \cdot f$. RF: spread-spectrum, short clocks, guard traces, ferrites, slow logic.

3 · PROCESSOR BASICS

µP = CPU only; µC = CPU+peripherals. **Buses:** address uni-dir (CPU→mem); data bidir; control mixed. CISC: many instr/modes, mem-to-mem, von Neumann. RISC: few, fixed length, load/store, many regs, Harvard OK. Blocks: state machine, ALU, regs, SR(flags), PC, SP(LIFO), IR, BIU. Assembly not portable. Z80 1976 8-bit (8080-compat); 68000 1979 16/32, 3.6 µm.
2's complement: invert+1 ($5E→$A2); 8-bit −128…+127; 16-bit −32768…+32767. Overflow only if same-sign operands give result that doesn't fit (V flag).

68000 · REGISTERS & FLAGS

D0-D7 (B/W/L); A0-A6, A7=SP (user/supv banked); PC; SR → CCR **X N Z V C**. A-regs word/long only. 24-bit address (16 MB), 16-bit ext. bus, big-endian. $ hex, % bin, @ oct. Operand order **src,dst**. CMP src,dst computes dst−src. Align: word even, long ×4.

68000 · EFFECTIVE ADDRESS

mode	form	meaning
0 / 1	Dn / An	register direct
2	(An)	pointer in An
3	(An)+	use, then An += size (1/2/4)
4	−(An)	An −= size, then use. PUSH=MOVE x,−(SP); POP=MOVE (SP)+,x
5	d16(An)	An + sext(d16)
6	d8(An,Xn)	An + Xn + sext(d8) (arrays: Xn=row·cols·sz)
7	abs.W/L, #imm, PC-rel	abs.W reaches $0-$7FFF & $FF8000-FFFFFF; PC-rel = relocatable

EA field = 3-bit mode + 3-bit reg. MOVE.W D0,A1 illegal → MOVEA (no flags). #imm forms: ADDI SUBI CMPI ORI.

68000 · INSTRUCTIONS

`MOVE.sz s,d` copy (.B/.W/.L), src unchanged · `LEA ea,An` · `CLR.L (A0)+` · ADD/SUB/ADDQ · `AND/OR/EOR` bitwise on size only, ANDI etc. immediates; AND #$F keeps digit, #0 clears; `NOT` 1 operand. `ASL.B #3` $01→$08 (0 in, MSB→C,X); count >8 in Dn; memory shifts 1 bit, word.
ASCII→hex: −$30 (0-9), −$37 (A-F), −$57 (a-f); ASL 12/8/4/0, OR. Test bit n: AND mask, check Z.
Machine code: op-code word + ext words (hi word first): NOP $4E71, MOVE.B D0,D3 $1600, ADD.L D4,D7 $DE84, ADDI.W #$4000,D5 $0645 $4000.
Directives: ORG, EQU (const), SET (redefinable), REG, DC (data), DCB (block), DS (reserve), END (start), OPT, MACRO.

Branch after CMP (signed / unsigned)		
D1<D0	BLT	BCS
D1≤D0	BLE	BLS
=	BEQ(Z=1)	BEQ
≠	BNE	BNE
D1>D0	BGT	BHI
D1≥D0	BGE	BCC (BHS)

BMI N=1, BPL N=0. Addresses unsigned → BHS. FOR/WHILE: test top, BRA bottom. DO-WHILE: body first. **JSR** pushes 32-bit ret addr (SP−4), **RTS** pops. Init SP at RAM top. `MOVEM.L list,-(SP)` … `(SP)+,list`. RTE returns from exception.
Zero block: LEA st,A0; LEA end,A1; L: CMPA.L A1,A0; BEQ done; CLR.L (A0)+; BRA L. **Largest byte:** MOVE.B (A0)+,D1; CMP.B D1,D0; BGE keep; MOVE.B D1,D0. **Smallest (HW3):** signed start $7F/BGT; unsigned $FF/BHI. **Add longs:** MOVE.L a,D0; ADD.L b,D0; MOVE.L D0,r. **Search $AA55:** CMP.W (A0)+,D0; BNE; LEA −2(A0),A2 (A2=0 if none).
HW2 trace: $4F then $17 via (A0)+ → word $4F17; AND.W $5555 → **$4515** (N=0 Z=0 V=C=0).

Z80 · ESSENTIALS

A,F; B C D E H L (pairs BC DE HL); alt set (EXX, EX AF,AF'); IX IY; SP (grows down); PC; I R. Flags S Z H P/V N C (H half-carry, N sub). Hex ends in **H**, leading 0 if A-F (0FFFFH). Order **dest,src** (LD B,A). Opcode byte(s)+imm: ADD A,22H = C6 22; 16-bit stored **little-endian**. 16 address lines (64 KB); paging: write 0-15 to a register to switch 64 KB pages (16 pages, 1 MB).
Addr: LD A,B · LD C,55H · LD DE,0A34BH · LD B,[HL] · LD [0FF20H],A (A only) · LD [nn],HL.
JP/JP cc, **JR** ±128, CALL/CALL cc, RET/RET cc, RETI; CP r; **DJNZ** (B−1≠0); BIT n,r (Z=~bit).
PUSH/POP pairs (2 B); CALL pushes 16-bit PC.
Directives: DEFB/.DB, DEFW, DEFS/.DS; Oshonsoft uses leading period. **No auto-inc, no 16-bit CP**: compare H then L. Largest: CP [HL]; JR NC (unsigned carry test).

68000 vs Z80 (quick compare)

	68000	Z80
operand order	src,dst	dst,src
hex	$FF	0FFH
address	24 b, 16 MB	16 b, 64 KB
endian	big	little (16-bit)
stack	−(SP)/(SP)+, JSR 32-bit	PUSH/POP pair, CALL 16-bit
return	RTS / RTE	RET / RETI
rel. jump	BRA ±32K, Bcc	JR ±128
loop	CMP+Bcc	DJNZ (B)
flags	X N Z V C	S Z H P/V N C
prefetch	yes (queue)	no
ASM directives	DC DS DCB	DEFB DEFW DEFS

EXAM TRAPS

- .W add on longwords adds only the low word.
- CMP src,dst: result is dst−src; signed BGE/BLT vs unsigned BCC/BCS.
- (An)+ moves by operand size (1/2/4), not always 1.
- Simulator shows bytes: word $4515 = $45 then $15.
- Page count uses chip depth, not width; chips = pages × (bus ÷ chip width).
- Wait-state math: use **3 clocks** (or the stated count), not 1.

ASSEMBLY LINE

LABEL (col 1) · OPCODE · OPERANDS (comma, no spaces) · comment (* or ;). Flow chart: □ op, ◇ decision. Reasons: bring-up, 1:1 with machine code, exact timing, no compiler.

MIDTERM PRACTICE · ANSWERS

P1	ρ(77 K)=5.67e-9·77+1.58e-8=4.524e-7 Ω·cm; R=0.1 V/8 A=0.0125 Ω; L=4 in=10.16 cm, t=0.002 in=0.00508 cm; A=ρL/R=3.677e-4 cm²; W=A/t=0.0724 cm = **0.0285 in (28.5 mil)**
P2	a F (addr is one-way) · b F (2 ns = **500 MHz**) · c ~T (both zero an 8-bit reg; CLR.B sets Z, LD C,0 no flags) · d F (66.7%) · e F (neg-logic OR = AND) · f T (2^{24}) · g T · h T (2^{30})
P3	24-bit addr, 8-bit bus, 256K×4: **2** chips/page; **128** chips; offset **A0-A17**, page **A18-A23**; **64** pages; pages $000000-03FFFF, $040000-07FFFF, $080000-0BFFFF, $0C0000-0FFFFF
P4	**Skew:** clock edge arrives at different parts at different times (path length, buffers) → less margin. **Split GND:** noisy digital vs quiet analog, joined at ONE point. **Bypass C:** local charge for fast edges, ~1 per 1-2 ICs (0.1 µF) + 10 µF bulk
P5	BEQ↔JR Z,foo · CMP.B D0,D1↔CP D · MOVE.B #$55,D7↔LD C,0A4H · MOVE.B D1,$FF35↔LD [0A3E7H],A · MOVE.B D6,D5↔LD B,A
P6	MOVE.W #$3400,(A3)+ = **36FC 3400** · ADD.B D6,$A4578A04 = **DD39 A457 8A04** · EOR.L D3,D6 = **B786** · SWAP D6 = **4846** · CLR.W $AD459BC0 = **4279 AD45 9BC0**

P7 count 1 bits (memory untouched): `MOVE.L $2000,D0` · `CLR.L D1` · `MOVE.W #32,D2` · loop: `LSL.L #1,D0` (top bit→C) · `BCC skip` · `ADDQ.B #1,D1` · skip: `SUBQ.W #1,D2` · `BNE loop` · `SIMHALT`. (Alt: CLR.L D3; LSL.L #1,D0; ADDX.L D3,D1; DBRA D2,loop with D2=31.) $F0F0A5C3 → 16.

Encoding recipe: MOVE 00|size(W=11)|dstReg|dstMode|srcMode|srcReg; ADD 1101|Dn|opmode|EA; EOR 1011|Dn|opmode|EA; abs.long = mode 111 reg 001; immediate = 111/100; SWAP = $4840+n.

4 · MEMORY DECODING

2^n locations for n lines. Chip 2^k×w: **offset** = A0…A(k–1) → chip pins; **page bits** = the rest → decoder. Pages = $2^{\text{page bits}}$. Chips/page = bus width ÷ w. Page start = n·2^k, end = start+2^k–1. ~CE = NAND(page bits, inverted where 0, **address-valid**). Page bits spread over A23…: e.g. $1A4 = 1 1010 0100.

Ex: 24-bit, 8-bit bus, 256K×4 → 18 offset, 6 page bits… (2 chips/page). 68000: A0-A14 offset, A15-A23 page: 512 pages of 32 KB. 24 lines: 16 MB. **74x138:** on if EN1=1, ~EN2=~EN3=0; ABC→Y0-Y7 (active-low). 3 page bits=8 regions (2 MB each, 16 MB). ~AS on ~EN2.

HW4 answers	
P1	20-bit: ROM 128K (A0-16): A19-17 = 000 / 001; RAM 64K (A0-15): A19-16 = 1000 / 1001; each ~CE = NAND(pattern,valid)
P2	A2-A31, D0-D31, 2M×8: space 4 GB; page = 4 chips (8 MB); **512 pages, 2048 chips**; chip A0-20 ← A2-22; page bits A23-31; pg 0 = 000000000, last = 111111111; ~WE0-3 per lane
P3	3T ≥ gates·2 ns + 15 ns. 2 gates → 19 ns → **157.9 MHz**; 3 gates → 21 ns → 142.9 MHz
P4	Page bits A20-23 (1 MB pg). Both 138: ABC=A20,21,22, ~EN2=~AS. A23→~EN3 of #1 (pg 0-7) and →EN1 of #2 (pg 8-15)

Memory design steps: chips/page → offset/page split → which pages present → decode uniquely. Byte-write strobes WR0..: byte=1, word=2, long=4 (A0,A1 implied). Alignment: word even, long ×4. 68000 map: vectors at 0, ROM, I/O, empty, RAM. **Timing:** data valid = decoder delay + t_{access} (from ~CE low) must fit in processor read window else add **wait states** (1 clock each) or faster part. Any gate cuts margin (74ALS138 up to 36 ns). Z80 sync, ~WAIT stretches.

Bus: plain outputs short; open-collector (pull-up, slow rise); **tri-state** (~OE). Drive bus only if ~CE=0 & ~OE=0. Active-low: ~NAME true when low; neg-logic OR = AND (De Morgan). 6T SRAM = 2 cross-coupled inverters + 2 pass.

SRAM: 5-6T/bit, no refresh, fast, cache. **DRAM:** 1T+C, refresh ~10-16 ms, row then column mux: 64 Mbit = 8192² = 13+13 bits; refresh every T/rows (≈2 µs). Read: row, ~RAS↓, col, ~WE=1, ~CAS↓, ~OE↓. Write: ~WE=0, data before ~CAS↓.

DMA: BUS REQ → CPU finishes cycle, tri-states, BUS GRANT → peripheral moves block → CPU resumes.

AFTER THE MIDTERM (final material)

5 · INTERRUPTS, I/O, STATE MACHINES

Interrupt async → ISR; polling wastes CPU (5k char/s UART @1 GHz ≈ 100k instr/char). Priority preempts; **NMI** unmaskable (68K level 7); RESET restarts; exceptions (÷0, illegal) unmaskable. **68000 vectors:** $000-$3FF, 256 long words; RESET: SP←$0, PC←$4; NMI at $7C; interrupt pushes SR+PC, supervisor mode, vector fetch. **I/O port:** DDR 1=out 0=in; read≠write (volatile). UART: poll DR before read, TB=0 before write.

S-R latch: S=R=1 undefined. FSM n-bit register → 2^n states; vending: S0→nickel S1→nickel unlock, dime unlock. **Moore** out=f(state), on nodes, clocked; **Mealy** out=f(state,input), on edges, fewer states, can glitch.

6 · TRANSISTORS & DIODES

Diode V_f≈0.6 V; LED R=(V_s–V_f)/I (V_f 1-4 V); Zener = reverse reference w/ series R.

BJT I_C=β·I_B, I_E=I_C+I_B, V_{be}≈0.6. Base R: I_b=I_c/β_{min}, **R=(V_{IO}–0.6)/I_b**, round R down. Saturation: $I_{c,max}$=(V_{cc}–0.4)/R_L; if β·I_B ≥ that → V_c=0.4 V, else V_c=V_{cc}–$I_C R_L$. Pick V_{max} 2×, I_{max} 2×, P=0.4·I, min β.

Open-collector: sinks only, pull-up (own voltage). ULN2003: 7 × 500 mA, 50 V. **Darlington** β=β1β2, on 1.2 V; **Sziklai** 0.6 V; R=(V_{IO}–V_{on})/(I_c/β). **PNP:** emitter to +rail, load to GND, sink base current R_x=(V_{rail}–0.6–V_{ol})/(I_{load}/β). **MOSFET:** voltage-controlled R, gate no current; N: load to +, P: load to GND (level shift); static-sensitive.

7 · DATA CONVERSION

Step = FS/(2^N–1): 10 V/255=0.04 V (8 b); 10 V/65535 (16 b). 10-bit 5 V: 4.89 mV. Fill range: amplify first.

Comparator out=1 if V+>V–. **Flash:** 2^N–1 comparators, thermometer→encoder, ~1 GSPS, 8-10 b, costly. **Single ramp:** counter+DAC, time ∝ V (≤2^N). **SAR:** binary search MSB first, N steps, 12-18 b, most common. **Dual slope:** V_x·T=V_{ref}·t_{ref}, 16-24 b, slow. **V/F:** F=k·V, counts=kVT, noise immune, slowest. **DAC:** binary currents i,2i,4i… summed; V=code·iR; FS=(2^N–1)iR.

Chain: condition → mux → sample&hold → ADC. Analog switch ~250 ns, ~100 Ω; mech ~10 ms bounce. **Droop:** t < (LSB/2)/droop (10 mV, 1 V/ms → 5 µs). **Aliasing:** sample ≥ 2·f_{max} (Nyquist) + low-pass filter. Furnace: 2 V range, 0.1° = 0.04 mV → 50× finer → 16 bit.

Schmitt: thresholds V_L/V_H, deadband. Debounce = RC + Schmitt. **Opto-isolator** ~1000 V, separate grounds. **Watchdog:** retrigger in time or RESET/NMI (RESET more reliable).

8 · SPECS, PARTITIONING, COST

HW: 10-100× faster, but NRE ≈$100K, ~3 mo, ASIC 50% first pass. SW: no BOM, field-fixable, slower, schedule risk. **Total = NRE + unit·volume**. HW $525K + $14/u (1k units = $539K) or $6/u (500k = $3.525M); SW $300K flat.

9 · DEBUG TOOLS

Monitor ($0-1K, not real-time, serial, run control). ROM emulator ($1-5K, real-time trace). **ICE** (~$20K): run control (BREAK→NMI), overlay memory, real-time trace. Prefetch/cache hide flow → bond-out, BDM, JTAG 1149.1.

Logic analyzer: timing mode internal clock, ±1 sample; record = rate·depth (4 ns × 4096 = 16.4 µs); glitch = >1 crossing. **State** mode on processor clock, time-stamped. Triggers: pattern (0101 0101 = 55H), edge; circular buffer shows pre-trigger (negative time); sequence levels; selective storage (start/stop addr).

10 · LABS

Lab 1 PRNG: D-FF shift reg + XORs (odd parity); 12 FF, 8 XOR; state mode, GND, D0-D7, CLK1, check T; timing 500 MHz (2 ns); crit path FF+XOR+setup. **Lab 2 Z80:** 4 MHz Z80A, 27C256 0000-7FFF, 62256 8000-FFFF, DS1232. **M1 cycle** T1-T4 (250 ns each @4 MHz): T1 addr, ~M1↓; ~MREQ,~RD↓; data read at T3 rise; T3-T4 refresh. Lengths: 09H 1 B, 3EH nn 2 B, 21H lo hi 3 B.

Lab 3 68008: 8 MHz, 8-bit bus, A0-A19, ROM 0-7FFF, RAM 8000-FFFF, GAL22V10; after RESET SP←[0], PC←[4]; 2-word **prefetch** (traces out of order; writes after next prefetch). ~CSROM: ~AS, A15=0; ~CSRAM A15=1; ~RD/~WR from R/~W. Bus cycle S0-S7 (4 clk); **~DTACK** sampled in S4: high → wait (+1 clk); 125 ns @8 MHz. ESD: wrist strap, ROMs in foam, power analyzer first.

HW1: schematic first: value, rating, tolerance, package, footprint; inst-amp gain 100-1000, ICL7660 –5 V, OP-07 ±3…±18 V, matched R; Gerber files (KiCad/Eagle).